GOV•UK

Department
for Environment,
Food & Rural Affairs

Guidance

Recyclability assessment methodology: assessing materials

Updated 4 September 2025

Contents

This publication is available at https://www.gov.uk/government/publications/recycling-assessment-methodology-materials-and-outputs/recycling-assessment-methodology-assessing-materials

You should read this guide alongside other RAM information that covers:

- how to assess your packaging waste (https://www.gov.uk/guidance/recycling-assessment-methodology-how-to-assess-your-packaging-waste)
- definitions and background (https://www.gov.uk/government/publications/recycling-assessment-methodology-background-and-definitions) - this explains the stages of recyclability and some technical terms
- clarifications and additional information (https://www.gov.uk/guidance/recyclability-assessment-methodology-supplementary-guidance) to help you carry out your RAM assessment
- the RAM decision tree (https://defracollectionandpackagingreform.createsend1.com/t/t-i-suytduy-l-i/)

Paper and board

Classification

Common examples of paper / board in packaging could include:

- cardboard boxes and sleeves
- corrugated cardboard, for example shipping packaging
- paperboard, for example cereal boxes and tissues boxes
- flexible paper packaging, for example wrappers and pouches
- moulded fibre, for example egg boxes

Paper and board components should consist mainly of natural fibres. They can include filling material, starch, clay or colour coatings including binder, as well as additives typically used in the paper industry such as wet-strength agents, sizing agents, dyes and bound water.

The paper and board waste stream may include fibre-based composites.

Fibre-based composite packaging with plastic content less than or equal to 5% by mass should be assessed through paper and board material guidance.

Fibre-based composite packaging with plastic content more than 5% by mass should be assessed through the fibre-based composite materials guidance.

Paper content can be determined by:

```
Paper content (w%)= (fibre+filler+water+additive+colour coating)(gsm)/total mass (gsm)
```

Where:

- fibre: is wood-based cellulose fibres
- filler: is mineral typical used in the industry, for example CaCO3 or kaolin
- additive: chemicals used for the paper making process, for example sizing agents, dry or wet strength agents, retention aids, defoamers, dyes, and pigments
- colour coating: mineralised coating layer used to enhance the brightness and printability of the paper
- total mass: total mass of the packaging product excluding separatable components which should be evaluated as an individual component

If the packaging unit or component uses alternatives to cellulosic wood derived fibres you must provide evidence that they are appropriate for use in papermaking and can be reprocessed without causing challenges in the processes or affecting other packaging waste materials. Industry standard testing methodologies may be used to supply this evidence.

Paper and board: collection

Paper and board is widely collected by 100% of local authorities and therefore meets the widely collected at kerbside criteria.

However, none of these variants of paper and board are widely collected at kerbside:

- fibre-based composite which has layers of plastic on both sides (double-sided lamination)
- paper and board to which glitter has been adhered
- greaseproof, siliconised or waxed paper
- parchment paper, for example baking paper
- padded polyethylene lined envelopes (unless easily separated by hand)

The items of packaging and components listed above should be classified as **red** unless there are dedicated take-back schemes available that meet the specified criteria of the take-back protocol. Items of packaging and components collected through a take-back scheme may qualify for **amber**, provided they continue to meet the conditions for subsequent recyclability stages.

Paper and board: sortation

The item of packaging or component must be at least 40mm in at least two dimensions to pass the sortation stage of the assessment.

Items of packaging and components smaller than 40mm in at least two dimensions can progress if one of the following applies:

- the undersized component can be attached to another component within the packaging unit by the consumer, whereby the resulting combined components meet the size threshold outlined and continue the assessment as one component
- there are take-back schemes available for undersized items that meet the criteria outlined in the take-back protocol. Undersized items collected through a take-back scheme may qualify for **amber**, provided they continue to meet the conditions for the next recyclability stages.

It is classed as **red** if it does not meet these criteria.

Paper and board: reprocessing

The paper and board reprocessing can be significantly impacted by some contaminating items.

If the item of packaging or component contains non-paper content greater than 15% by weight and not classified as a fibre-based composite, it is **red**.

If any of the following fillers, additives, or agents have been intentionally added to the item of packaging or component, it is **red**:

- urea/formaldehyde
- urea/melamine

If will not be **red** if you can provide test results to confirm that it can be reprocessed without causing disruptions to the processes or affecting other packaging waste materials.

If glass or carbon fibres have been intentionally added to the item of packaging or component, it is **red**.

If the item of packaging or component contains any of the following laminations or coatings, it is **red**:

- two-sided wax coating, for example molten wax dip coated, but does not apply to waxes used in printing inks

If the item of packaging or component contains any of the following other contaminants, it is **red**:

- siliconising agents, for example used in papers for labels

Update for version 1.1
The following requirement has been removed from the RAM for 2025 assessment:

If the item of packaging or component contains any retained product residue that cannot be removed by hand, it is **red**. A common example would be 3D food baked on and attached to the paper or board. Surface staining such as small oil marks that do not fully soak the paper, or crumbs can be tolerated.

Paper and board: application

The quality of reprocessing output ('recyclate') can be affected in standard paper milling facilities by some contaminating items.

If the item of packaging or component contains any of the following materials, it is **amber**:

- non-paper content greater than 10% by weight and not classified as a fibre-based composite
- non-wood-based fibres, for example bagasse, palm, fibre, rice straw, wheat straw, barley straw, oat straw, grass straw, flax, hemp, and bamboo, unless test results are provided to confirm that it can be reprocessed without causing disruptions to the processes or affecting other packaging waste materials

If the item of packaging or component contains any of the following adhesives, it is **amber**:

- adhesive lamination (inside of pack) of PET, mPET, PET/PE, unless test results are provided to confirm that it can be reprocessed without causing disruptions to the processes or affecting other packaging waste materials.

If the item of packaging or component contains any of the following laminations or coatings, it is **amber**:

- pVDC or PVC polymer dispersion coatings, unless test results are provided to confirm that it can be reprocessed without causing disruptions to the processes or affecting other packaging waste materials.
- lamination with aluminium foil where the coating thickness is greater than or equal to 6 micron (μm)
- wax dispersion, including microcrystalline waxes but does not apply to waxes used in printing inks

If the item of packaging or component contains any of the following barrier metallisation, it is **amber**:

- direct metallisation, including primer, aluminium nanoscale, or protective coating,
- transfer metallisation, including adhesive and transfer metallisation

Unless test results are provided to confirm that it can be reprocessed without causing disruptions to the processes or affecting other packaging waste materials.

If a packaging item or component does not contain any of the above and passed through the previous stages it is classed as **green**.

Update for version 1.1
The following requirement has been removed from the RAM for 2025 assessment:

If the item of packaging or component contains any of the following additives, it is **amber**:

* Polyamid epichlorohydrin (PAE), unless test results are provided to confirm that it can be reprocessed without causing disruptions to the processes or affecting other packaging waste materials.

If the item of packaging or component contains any of the following inks and varnishes, it is **amber**:

- Ultra Violet (UV) cured varnish greater than 4g/m2 with 100% coverage, unless test results are provided to confirm that it can be reprocessed without causing disruptions to the processes or affecting other packaging waste materials.
- Inks containing mineral oils where they have been intentionally added

Fibre-based composite materials

Fibre-based composite materials: classification

Common examples of fibre-based composite materials in packaging could include:

- liquid food and drink cartons
- sandwich skillets
- food boxes and trays
- cardboard boxes and sleeves
- flexible paper packaging, for example wrappers and pouches

Fibre-based composite packaging with plastic content less than or equal to 5% by mass should be assessed through paper and board material guidance.

Fibre-based composite packaging with plastic content greater than 5% by mass should be assessed through fibre-based composite materials guidance.

Paper content can be determined by:

```
Paper content (w%)= (fibre+filler+water+additive+colour coating)(gsm)/total mass (gsm)
```

Where:

- fibre: is wood-based cellulose fibres
- filler: is mineral typical used in the industry, for example CaCO3 or kaolin
- additive: chemicals used for the paper making process, for example sizing agents, dry or wet strength agents, retention aids, defoamers, dyes, and pigments
- colour coating: mineralised coating layer used to enhance the brightness and printability of the paper
- total mass: total mass of the packaging product excluding separatable components which should be evaluated as an individual component

If the packaging unit or component uses alternatives to cellulosic wood derived fibres you must provide evidence that they are appropriate for use in papermaking and can be reprocessed without causing challenges in the processes or affecting other packaging waste materials. Industry standard testing methodologies may be used to supply this evidence.

If your fibre-based composite packaging item is a liquid carton, refer to the guidance directly below for fibre-based composite materials (liquid cartons). For fibre-based composite packaging items that do not fall into this category, proceed to the section on fibre-based composite materials (non-liquid cartons).

Fibre-based composite materials (liquid cartons): collection

To meet the widely collected at kerbside criteria, an item of packaging or component must be collected by 75% of local authorities. Items of packaging and components may progress via the limited collections route if collected by at least 50% of local authorities but are capped at **amber**, unless otherwise specified.

For fibre-based composites:

- liquid food and drink cartons (fibre-based composite) are collected by 66% of local authorities

Fibre-based composite (liquid cartons) packaging not meeting the description above should be classified as **red** unless there are dedicated take-back schemes available that meet the specified criteria of the take-back protocol. Items of packaging and components collected through a take-back scheme may qualify for **amber**, provided they continue to meet the conditions for subsequent recyclability stages.

Fibre-based composite materials (liquid cartons): sortation

The item of packaging or component must be at least 40mm in at least 2 dimensions to pass the sortation stage of the assessment.

Items of packaging and components smaller than 40mm in at least 2 dimensions can progress if one of the following applies:

- the undersized component can be attached to another component within the packaging unit by the consumer, whereby the resulting combined components meet the size threshold outlined and continue the assessment as one component
- there are take-back schemes available for undersized items that meet the criteria outlined in the take-back protocol. Undersized items collected through a take-back scheme may qualify for an **amber**, provided they continue to meet the conditions for subsequent recyclability stages.

It is classed as **red** if it does not meet these criteria.

Fibre-based composite packaging collected through dedicated take-back schemes and separated at source will pass the sortation stage of the assessment. These items may qualify for an **amber** classification, provided they continue to meet the requirements for subsequent stages.

Liquid food and drink cartons with any outer layer, other than PE or paper, should be classed as **red** unless evidence and testing demonstrate that they can be reliably identified by near-infrared (NIR) sensor-based sortation systems.

Fibre-based composite materials (liquid cartons): reprocessing

The reprocessing of fibre-based composites will occur at specialist mills equipped to handle a mixture of materials, including paper, aluminium, and polyethylene (PE).

If the item of packaging or component contains any of the following materials, it is **red**:

- polyethylene (PE) with less than 80% by weight of polymer content
- polypropylene (PP) exceeding 20% by weight by polymer content
- polyethylene terephthalate (PET) exceeding 5% by weight of polymer content
- biodegradable polymers in any proportion of polymer content

Update for version 1.1
The following requirement has been removed from the RAM for 2025 assessment:

Packaging that is likely to retain product residue that a consumer cannot remove by hand is **red**. A common example would be food attached to fibre-based composite packaging.

Fibre-based composite materials (liquid cartons): application

The quality of reprocessing output ('recyclate') can be affected by some contaminating items. The item of packaging or component is **amber** if it contains any of the following:

If the item of packaging or component contains any of the following, it is **amber**:

- polyethylene (PE) between 80% and 90% by weight of polymer content
- polypropylene (PP) between 10% and 20% by weight by polymer content
- polyethylene terephthalate (PET) with less than 5% by weight of polymer content

If the item of packaging or component contains any of the following coatings, it is **amber**:

- wax coatings, including wax emulsions and dispersions, but does not apply to waxes used in printing inks

If any of the following fillers, additives or agents have been intentionally added to the item of packaging or component contains , it is **red**:

- urea/formaldehyde, unless you can provide test results to confirm that it can be reprocessed without causing disruptions to the processes or affecting other packaging waste materials.

If an item of packaging or component does not contain any of the above and has met the conditions outlined in prior stages it is classed as **amber** due to the current collection constraints.

If your fibre-based composite packaging item is not a liquid food and drink carton, refer to the guidance directly below for fibre-based composite materials (non-liquid cartons).

Update for version 1.1
The following requirement has been removed from the RAM for 2025 assessment:

If the item of packaging or component contains any of the following fillers, additives or agents intentionally added, it is **amber**:

- Polyamid epicholorohydrin (PAE)

Fibre-based composite materials (non-liquid cartons): collection

Fibre-based composite packaging, which are not considered liquid food and drink cartons, are widely collected by 100% of local authorities and therefore meets the widely collected at kerbside criteria.

However, none of these variants of paper and board are widely collected at kerbside:

- fibre-based composite packaging with more than 15% non-paper content by weight
- fibre-based composite which has layers of plastic on both sides (double-sided lamination)
- paper and board to which glitter has been adhered
- greaseproof, siliconised or waxed paper
- parchment paper, for example baking paper
- padded polyethylene lined envelopes (unless easily separated by hand)

The items of packaging and components listed above should be classified as **red** unless there are dedicated take-back schemes available that meet the specified criteria of the take-back protocol. Items of packaging and components collected through a take-back scheme may qualify for **amber**, provided they continue to meet the conditions for subsequent recyclability stages.

Fibre-based composite materials (non-liquid cartons): sortation

The item of packaging or component must be at least 40mm in at least 2 dimensions to pass the sortation stage of the assessment.

Items of packaging and components smaller than 40mm in at least 2 dimensions can progress if one of the following apply:

- the undersized component can be attached to another component within the packaging unit by the consumer, whereby the resulting combined components meet the size threshold outlined and continue the assessment as one component
- there are take-back schemes available for undersized items that meet the criteria outlined in the take-back protocol.

Undersized items collected through a take-back scheme may qualify for **amber**, provided they continue to meet the conditions for reprocessing and application

Fibre-based composite materials (non-liquid cartons): reprocessing

The fibre-based composite reprocessing can be significantly impacted by some contaminating items.

If any of the following fillers, additives or agents have been intentionally added to the item of packaging or component contains , it is **red**:

- urea/formaldehyde
- urea/melamine

If test results are provided to confirm that it can be reprocessed without causing disruptions to the processes or affecting other packaging waste materials, it is not **red**.

If the item of packaging or component contains any of the following fibres, it is **red**:

- glass or carbon fibres

If the item of packaging or component contains any of the following laminations or coatings, it is **red**:

- two-sided lamination, for example PE/paper/PE, PP/paper/PP, PET/paper/PET, unless there is clear consumer guidance for peeling off the lamination
- two-sided wax coating, for example molten wax dip coated, but does not

apply to waxes used in printing inks

If the item of packaging or component contains any of the following other contaminants, it is **red**:

- siliconising agents, for example used in papers for labels

Update for version 1.1
The following requirement has been removed from the RAM for 2025 assessment:

- If the item of packaging or component contains any retained product residue that cannot be removed by hand, it is **red**. A common example would be 3D food baked on and attached to the paper or board. Surface staining such as small oil marks that do not fully soak the paper, or crumbs can be tolerated.

Fibre-based composite materials (non-liquid cartons): application

The quality reprocessing output ('recyclate') can be affected in standard paper milling facilities by some contaminating items.

If the item of packaging or component contains any of the following materials, it is **amber**:

- non-paper content greater than 10% by weight
- non-wood-based fibres, for example bagasse, palm, fibre, rice straw, wheat straw, barley straw, oat straw, grass straw, flax, hemp, and bamboo, unless test results are provided to confirm that it can be reprocessed without causing disruptions to the processes or affecting other packaging waste materials.

If the item of packaging or component contains any of the following adhesives, it is **amber**:

- adhesive lamination (inside of pack) of PET, mPET, PET/PE, unless test results are provided to confirm that it can be reprocessed without causing disruptions to the processes or affecting other packaging waste materials.

If the item of packaging or component contains any of the following laminations or coatings, it is **amber**:

- PVDC / PVC polymer dispersion coatings, unless test results are provided to confirm that it can be reprocessed without causing disruptions to the processes or affecting other packaging waste materials.
- lamination with Aluminium foil where the coating thickness is greater than or equal to 6 micron (μm)
- wax dispersion, including microcrystalline waxes but does not apply to waxes used in printing inks

If the item of packaging or component contains any of the following barrier metallisation, it is **amber**:

- direct metallisation, including primer, aluminium nanoscale, or protective coating,
- transfer metallisation, including adhesive and transfer metallisation

If test results are provided to confirm that it can be reprocessed without causing disruptions to the processes or affecting other packaging waste materials, it is not **amber**.

If a packaging item or component does not contain any of the above and passed through the previous stages it is classed as **green**.

Update for version 1.1
The following requirement has been removed from the RAM for 2025 assessment:

If the item of packaging or component contains any of the following additives, it is **amber**:

- Polyamid-epichlorohydrin (PAE), unless test results are provided to confirm that it can be reprocessed without causing disruptions to the processes or affecting other packaging waste materials.

If the item of packaging or component contains any of the following inks and varnishes, it is **amber**:

- Ultra Violet (UV) cured varnish greater than 4g/m2 with 100% coverage, unless test results are provided to confirm that it can be reprocessed without causing disruptions to the processes or affecting other packaging waste materials.

- Inks containing mineral oils where they have been intentionally added

Plastic (flexibles)

Plastic (flexibles): classification

Plastics classified as 'flexible' refer to items that change shape when filled. Common examples of flexible plastic packaging could include:

- bags
- pouches
- sachets
- sleeves
- wrappers
- lidding film or liners
- crisp packets
- fruit nets

Plastic films can be made from various types of plastic polymers, including polyolefins (PO), polyethylene (PE), polypropylene (PP), and polyvinyl chloride (PVC) and also includes metallised films.

Plastic (flexibles): collection

To meet the widely collected at kerbside criteria, an item of packaging or component must be collected by 75% of local authorities. At present, no plastic (flexibles) exceeds a 14% collection rate and therefore fail to meet this criterion, or the criteria for the limited collection route.

Flexible plastic packaging types may progress via the take back route if a valid scheme is available but are capped at **amber**. It is the responsibility of the producer to prove they meet the criteria set out in the take back protocol. If you are not certain that you meet these criteria, contact your take back scheme provider.

If neither kerbside collections or the take back protocol apply, then the item of packaging or component is classified as **red**.

Plastic (flexibles): sortation

If the item of packaging or component contains any of the following pigments, it is **red**:

- Carbon black pigment within the masterbatch (this does not apply to inks or labels)

If the item of packaging or component contains any of the following materials, it is **red**:

- aluminium foil layers

Update for version 1.1
The following requirement has been removed from the RAM for 2025 assessment:

- Carbon black pigment in inks and labels covering more than 50% of the total surface area

Plastic (flexibles): reprocessing

Polyolefin-based plastic film packaging and plastic bags which contain a minimum of 80% by weight of polyethylene, polypropylene, or a combination of both that still equates to at least 80% of the total composition by weight, can be considered for reprocessing. Any item of packaging or component below this threshold is classified as **red**.

Plastic (flexibles) reprocessing can be significantly impacted by some contaminating items.

If the item of packaging or component contains any of the following materials, it is **red**:

- PET
- PVC
- PVDC
- non-PE and non-PP foamed polymer layers
- oxo-degradable, bio-degradable plastic, or compostable plastic
- paper
- aluminium foil - this does not include metalised films.
- ethylene-vinyl alcohol (EVOH) as barriers or coatings exceeding 10% of the total weight

If the item of packaging or component contains any of the following additives or fillers, it is **red**:

- oxo-degradability additives
- foamed thermoplastic non-polyolefin elastomers

If the item of packaging or component's density is greater than 1g/cm3, it is **red**.

If the item of packaging or component uses lacquers and inks containing PVC binders, it is **red**.

Plastic (flexibles): application

While technically capable of being recycled, an item of packaging or component containing anything listed below are more complex to reprocess, can reduce the quality of recyclate produced, or cause unnecessary secondary material loss and are therefore classified as **amber**.

- Attached labels or sleeves of a different material type

An item of packaging or component is also classified as **amber** if any of the following apply:

- adhesives such as polyurethane exceeding 3% of the total component weight when applied to PE

- adhesives such as polyurethane exceeding 5% of the total component weight when applied to PP
- adhesives such as acrylic or natural rubber latex adhesives, as well as non-PE or non-PP based tie layers exceeding 5% of the total component weight

If an item of packaging or component meets the conditions above and has met the conditions outlined in prior stages it is classed as **amber** due to the current collection constraints.

Plastic (rigids)

Plastic (rigids): classification

Plastics classified as “rigid” in packaging refer to items that maintain a defined shape and structural integrity under normal conditions of use. Common examples of rigid plastic packaging could include:

- bottles
- pots
- tubs
- trays
- tubes
- caps and closures

Plastic (rigids): collection

To meet the widely collected at kerbside criteria, an item of packaging or component must be collected by 75% of local authorities. For plastic (rigids) this includes:

- plastic bottles collected by 100% of local authorities
- rigid mixed plastics (pots, tubs and trays) collected by 88% of local authorities

Plastic rigid items of packaging and components, not listed above, may progress via the limited collections route if collected by at least 50% of local authorities but are capped at **amber**.

Items of packaging and components that do not meet the wide or limited collection criteria, may progress via the take back route if a valid scheme is available but are capped at an **amber** rating. It is the responsibility of the producer to prove they meet the criteria set out in the take back protocol. If you are not certain that you meet these criteria, contact your take back scheme provider.

If neither kerbside collections or the take back protocol apply, then the item of packaging or component is classified as **red**.

Plastic (rigids): sortation

If the item of packaging or component uses a masterbatch containing carbon black pigment, it is **red**.

The item of packaging or component must also be at least 40mm in at least 2 dimensions to pass the sortation stage of the assessment. Components smaller than this can progress if one of the following applies:

- the undersized component can be attached to another component within the packaging unit by the consumer, whereby the resulting combined components meet the size threshold outlined and continue the assessment as one component
- there are take-back schemes available that meet the criteria outlined in the take-back protocol. Packaging units and components collected through a take-back scheme may qualify for **amber**, provided they continue to meet the conditions for subsequent stages

Update for version 1.1
The following requirement has been removed from the RAM for 2025 assessment:

If the item of packaging or component has an attached label or sleeve of a different material or polymer type covering more than the relevant threshold outlined below, it is **red**.

- for bottles an attached label or sleeve should not exceed 40% of the total surface area
- for pots, tubs, and trays an attached label or sleeve should not exceed 60% of the total surface

Carbon black pigment in inks and labels covering more than 50% of the total surface area.

Plastic (rigids): reprocessing

Plastic (rigids) reprocessing can be significantly impacted by some contaminating items therefore if an item of packaging or component contains any of the following it is classified as **red**:

- PVC (including non-PVC with PVC components)
- polystyrene (including but not limited to HIPS, expanded & extruded)
- oxo-degradable, biodegradable or compostable plastics
- non-polyolefin foamed plastics e.g. non-PP and non-PE

Different polymers require distinct reprocessing techniques and therefore specific contaminants in addition to those above are listed below.

Polyethylene terephthalate (PET) bottle items of packaging and components are classified as **red** if they contain:

- ethylene-vinyl alcohol (EVOH) as a barrier or coating exceeding 10% of the total weight
- caps or seals comprised of steel or aluminium with a density greater than or equal to 1g/cm3
- caps, seals or valves comprised of silicone
- attached labels or sleeves that are PVC, metallised, or PS with a density greater than 1g/cm3

Polyethylene terephthalate (PET) tray items of packaging and components are classified as **red** if they contain:

- EVOH as a barrier or coating exceeding 10% of the total weight
- attached labels or sleeves that are PET, PVC, or metallised or PS with a density greater than 1g/cm3

High-density polyethylene (HDPE) items of packaging and components are classified as **red** if they contain:

- PVDC barriers or coatings
- caps comprised of steel, aluminium, PS, PVC, or thermoset plastics
- liners comprised of PS, PVC, or EVA with aluminium
- seals comprised of PVC or silicone
- attached labels or sleeves comprised of PVC, aluminium, metallised PET, or metallised PS

Polypropylene (PP) items of packaging and components are classified as **red** if they contain:

- PVDC barriers or coatings
- caps comprised of steel, aluminium, PS, PVC, or thermoset plastics
- attached labels or sleeves comprised of PVC or metallised PET
- inserts comprised of PVC, PS, polyurethane (PU), PA (nylon), PET (heavy), polycarbonate (PC), acrylic (PMMA), thermoset plastics, or metallics

Update for version 1.1
The following requirement has been removed from the RAM for 2025 assessment:

- Attached label adhesives which are not removable in an 80°C hot wash

Polyethylene terephthalate (PET) bottle items of packaging and components are classified as **red** if they contain:

- Nanocomposite additives

Polyethylene terephthalate (PET) tray items of packaging and components are classified as **red** if they contain:

- PE seal layers

High-density polyethylene (HDPE) items of packaging and components are classified as **red** if they contain:

- Additives that increase the density of HDPE above 0.995 g/cm3 including Talc, CaCO3 and other fillers This applies to the plastic itself and does not apply to any inks used.

Plastic (rigids): application

While technically capable of being recycled, items of packaging and components containing anything listed below are more complex to reprocess, can reduce the quality of recyclate produced, or cause unnecessary secondary material loss and are therefore classified as **amber**:

- use of foil
- ethylene-vinyl alcohol (EVOH) exceeding 5% of the total weight

Different polymers require distinct reprocessing techniques and therefore specific contaminants in addition to those above are listed below.

Polyethylene terephthalate (PET) bottle items of packaging and components are classified as **amber** if they contain:

- any of the following colours: dark blue, dark green or brown - this applies to the plastic itself and does not apply to any inks used
- external coatings or PA-3 layers
- any of the following additives: UV stabilisers or AA blockers

Polyethylene terephthalate (PET) tray items of packaging and components are classified as **amber** if they contain:

- any of the following additives: O2 scavengers, UV stabilisers and AA blockers
- inserts comprised of HDPE, LDPE, PP, PET, or paper

High-density polyethylene (HDPE) items of packaging and components are classified as **amber** if they contain:

- any of the following colours: light blue, green, light tints or opaque colours - this applies to the plastic itself and does not apply to any inks used.
- polyamide (PA) including MXD6 as barriers or coatings
- seals comprised of aluminium

Polypropylene (PP) items of packaging and components are classified as **amber** if they contain:

- opaque colours, excluding white - this applies to the plastic itself and does not apply to any inks used
- polyamide (PA) including MXD6 as barriers or coatings
- inserts comprised of HDPE, LDPE, paper, or PET

If an item of packaging or component does not contain any of the above and has met the conditions outlined in prior stages it is classed as **green**.

Update for version 1.1
The following requirement has been removed from the RAM for 2025 assessment:

While technically capable of being recycled, items of packaging and components containing anything listed below are more complex to reprocess, can reduce the quality of recyclate produced, or cause unnecessary secondary material loss and are therefore classified as **amber**:

- Attached labels or sleeves comprised of paper

Polyethylene terephthalate (PET) bottle items of packaging and components are classified as **amber** if they contain:

- Any of the following colours: heavy colours

Polypropylene (PP) items of packaging and components are classified as **amber** if they contain:

- Heavy colours
- Clarifier additives
- Attached labels or sleeves comprised of paper
- Inserts comprised of PET (light)

Polypropylene (PP) items of packaging and components are classified as

amber if they contain:

- Attached labels or sleeves comprised of paper

Steel

Steel: classification

Common examples of steel in packaging could include:

- food tins
- aerosols
- personal care products
- decorative cans
- lids from glass bottles or jars
- rigid steel containers
- closures

Steel: collection

To meet the widely collected at kerbside criteria, an item of packaging or component must be collected by 75% of local authorities. For steel this includes:

- aerosols collected by 94% of local authorities
- food cans or tins collected by 100% of local authorities
- metal lids on glass jars collected with glass bottles and jars by 89% of local authorities
- foil and foil trays collected by 84% of local authorities

Items of packaging and component types, not listed above, may progress via the limited collections route if collected by at least 50% of local authorities but are capped at **amber**, unless otherwise specified.

Items of packaging and components that do not meet the wide or limited collection criteria, may progress via the take back route if a valid scheme is available but are capped at an **amber** rating. It is the responsibility of the producer to prove they meet the criteria set out in the take back protocol. If you are not certain that you meet these criteria, contact your take back scheme provider.

If neither kerbside collections or the take back protocol apply, then the item of packaging or component is classified as **red**.

Steel: sortation

In order to pass this stage of the assessment, the item of packaging or component must not exceed 300mm in height, width or length. If above this threshold a component will be classified as **red** unless one of the following applies:

- the oversized component can be dismantled or folded by the consumer, whereby the resulting broken down or folded components meet the size threshold outlined
- there are take-back schemes available that meet the criteria outlined in the take-back protocol. Packaging units and components collected through a take-back scheme may qualify for an **amber**, provided they continue to meet the conditions for subsequent stages

Steel: reprocessing

The steel reprocessing infrastructure in the UK is equipped to handle contaminants that can reasonably be expected to appear in an item of packaging, therefore if an item of packaging or component has made it to this stage of the assessment it should proceed to application.

Steel: application

While technically capable of being recycled, an item of packaging or component containing more than 30% non-steel content by weight is classified as **amber** as this causes unnecessary secondary material loss.

If the item of packaging or component contains less than 30% non-steel content by weight, it is classified as **green**.

Aluminium

Aluminium: classification

Common examples of aluminium packaging could include:

- food containers or tins
- lids from glass bottles or jars
- aluminium tubes
- bottles
- rigid containers
- closures
- aerosols
- foil trays
- laminated foils

Films metallised via vacuum deposition, including those used in crisp packets, should be assessed under the predominant material by weight.

Aluminium: collection

To meet the widely collected at kerbside criteria, an item of packaging or component must be collected by 75% of local authorities. For aluminium this includes:

- aerosols collected by 94% of local authorities
- cans / tins / bottles collected by 100% of local authorities
- metal lids and closures on glass bottles and jars collected by 89% of local authorities with glass bottles and jars
- foil and foil trays collected by 84% of local authorities

Packaging types, not listed above, may progress via the limited collections route if collected by at least 50% of local authorities but are capped at **amber**, unless otherwise specified.

Item of packaging or components that do not meet the wide or limited collection criteria, may progress via the take back route if a valid scheme is available but are capped at an **amber** rating. It is the responsibility of the producer to prove they meet the criteria set out in the take back protocol. If you are not certain that you meet these criteria, contact your take back scheme provider.

If neither kerbside collections or the take back protocol apply, then the item of packaging or component is classified as **red**.

Aluminium: sortation

In order to pass this stage of the assessment, the item of packaging or component must not exceed 300mm in any height, width or length. If above this threshold a component will be classified as **red** unless one of the following apply:

- the oversized component can be dismantled or folded by the consumer, whereby the resulting broken down or folded components meet the size threshold outlined

- there are take-back schemes available that meet the criteria outlined in the take-back protocol. Packaging units and components collected through a take-back scheme may qualify for an **amber**, provided they continue to meet the conditions for subsequent stages

Aluminium: reprocessing

The aluminium reprocessing infrastructure in the UK is well equipped to handle contaminants that can reasonably be expected to appear in an item of packaging, therefore if a component has made it to this stage of the assessment it should proceed to application.

Aluminium: application

While technically capable of being recycled, items of packaging and components containing more than 30% non-aluminium content by weight is classified as **amber** as this causes unnecessary secondary material loss.

If the item of packaging or component contains less than 30% non-aluminium content by weight, it is classified as **green**.

Glass

Glass: classification

Common examples of glass packaging could include:

- bottles
- jars

Glass: collection

Glass packaging is collected by 89% of local authorities and therefore meets the widely collected at kerbside criteria. However, none of these variants of glass are widely collected at kerbside:

- mirrored glass
- heat-resistant or lead glass
- decorative glass
- glass with designed-in contamination (meaning that product residue cannot be easily removed by the consumer, such as nail polish bottles or concealer)

Items of packaging and components that do not meet the kerbside collection criteria may progress via the take back route if a valid scheme is available but are capped at an **amber** rating. It is the responsibility of the producer to prove they meet the criteria set out in the take back protocol. If you are not certain that you meet these criteria, contact your take back scheme provider.

If neither kerbside collections or the take back protocol apply, then the item of packaging or component is classified as **red**.

Glass: sortation

Due to the nature of glass reprocessing, no sortation specifications are defined and therefore any item of packaging or component that has made it to this stage of the assessment should progress to reprocessing.

Glass: reprocessing

The glass reprocessing infrastructure in the UK is well equipped to handle contaminants that can reasonably be expected to appear in an item of packaging, therefore if an item of packaging or component has made it to this stage of the assessment it should proceed to application.

Glass: application

While technically capable of being recycled, packaging containing anything listed below are more complex to reprocess, can reduce the quality of recyclate produced, or cause unnecessary secondary material loss and are therefore classified as **amber**.

- ceramic swing stoppers
- non-glass attachments or inserts that cannot be separated by hand, other than attached labels, such as pumps or dispensers - this does not apply to metal attachments such as screw-top skirts or collars
- any colour other than clear (flint), green, blue, or amber (brown). this applies to the glass itself and does not apply to any inks used.

If none of the above apply, the item of packaging or component is classified as **green**.

Update for version 1.1
The following requirement has been removed from the RAM for 2025 assessment:

While technically capable of being recycled, packaging containing anything listed below are more complex to reprocess, can reduce the quality of recyclate produced, or cause unnecessary secondary material loss and are therefore classified as **amber**.

- Attached label covering more than 60% of the total surface area

Wood

Wood: classification

Wood has limited applications in household packaging, but some examples could include:

- decorative and novel components
- food trays where there is another material, such as fibre / paper sleeve for food contact
- wooden batons
- wooden pallets

Wood: collection

To meet the widely collected at kerbside criteria, an item of packaging or component must be collected by 75% of local authorities. At present, no wood packaging exceeds a 0% collection rate and therefore fail to meet this criterion, or the criteria for the limited collection route.

Wood packaging types may progress via the take back route if a valid scheme is available but are capped at an **amber** rating. It is the responsibility of the producer to prove they meet the criteria set out in the take back protocol. If you are not certain that you meet these criteria, contact your take back scheme provider.

If none of the above apply, then the component is classified as **red**.

Wood: sortation

Given wood packaging is not eligible for household collection, there is subsequently no sufficiently scaled process for the sortation of wood packaging components and therefore all wood is classified as **red**.

Wood: reprocessing

While wood is technically capable of being reprocessed, it is not practically reprocessed at scale within the household packaging recycling infrastructure as a result of it not being collected and / or sorted at a sufficient scale. Therefore, all wood is classified as **red**.

Wood: application

At present, no item of wood packaging is expected to meet this stage of the assessment and therefore no criteria for appropriate applications are defined and all wood is classified as **red**.

Other materials

Classification

If a producer is unable to ascertain which of the material categories their packaging falls into, or the packaging does not fall within the categories outlined, the packaging should be classified as ‘other’. Examples of other materials could include:

- cork
- bamboo
- ceramic
- copper
- hemp
- rubber
- silicone

Where the category is known but the technical specifications required for this assessment are not, the packaging should be categorised as **red** under the predominant material by weight.

Collection

To meet the widely collected at kerbside criteria, an item of packaging or component must be collected by 75% of Local Authorities. At present, no packaging known to be classified as ‘other’ exceeds a 0% collection rate and therefore fail to meet this criterion, or the criteria for the limited collection route.

‘Other’ packaging types may progress via the take back route if a valid scheme is available but are capped at an **amber** rating. It is the responsibility of the producer to prove they meet the criteria set out in the take back protocol. If you are not certain that you meet these criteria, contact your take back scheme provider.

If neither kerbside collections or the take back protocol apply, then the component is classified as **red**.

Sortation

Given ‘other’ packaging is not eligible for household collection, there is subsequently no sufficiently scaled process for the sortation of ‘other’ packaging components and therefore all ‘other’ materials are classified as **red**.

Reprocessing

While ‘other’ packaging may be technically capable of being reprocessed, it is not practically reprocessed at scale within the household packaging recycling infrastructure as a result of it not being collected and / or sorted at a sufficient scale. Therefore all ‘other’ is classified as **red**.

Application

At present, no item of ‘other’ packaging is expected to meet this stage of the recyclability assessment and therefore no criteria for appropriate applications are defined and all ‘other’ is classified as **red**.